

cardiac arrhythmia and torsades de pointes, have been seen in treatment with macrolides, including azithromycin. Prescribers should consider the risk of QT prolongation should be considered, which can be fatal when weighing the risks and benefits of azithromycin for at-risk groups including:

- Patients with congenital and documented QT prolongation.
- Patients currently receiving treatment with other active substances known to prolong QT interval, such as antiarrhythmics of Classes IA and III, antipsychotic agents, antidepressants, and fluoroquinolones.
- Patients with electrolyte disturbance, particularly in cases of hypokalemia and hypomagnesemia.
- Patients with clinically relevant bradycardia, cardiac arrhythmia, or cardiac insufficiency.
- Elderly patients, as they may be more susceptible to drug-associated effects on the QT interval.

Aspartame
Unsuitable for phenylketonurics as product contains aspartame.

Myasthenia gravis
Exacerbations of the symptoms of myasthenia gravis and new onset of myasthenia syndrome have been reported in patients receiving azithromycin therapy.

INTERACTIONS WITH OTHER MEDICAMENTS
Antacids
No effect on overall bioavailability was seen in simultaneous administration of antacid with azithromycin although peak serum concentrations were reduced by approximately 24%. In patients receiving both azithromycin and antacids, the medicinal products should not be taken simultaneously.

Cetirizine
Co-administration of a 5-day regimen of azithromycin with 20 mg of cetirizine at steady-state resulted in no pharmacokinetic interactions and no significant changes in QT interval.

Didanosine (Dideoxyinosine)
Co-administration of 1200 mg/day azithromycin with 400 mg/day didanosine did not affect the steady state pharmacokinetics of didanosine.

Digoxin and colchicine (P-gp substrates)
Concomitant administration of macrolide antibiotics, including azithromycin, with P-glycoprotein substrates such as digoxin and colchicine, has been reported to result in increased serum levels of the P-glycoprotein substrate. Therefore, if azithromycin and P-gp substrates such as digoxin are administered concomitantly, the possibility of elevated serum concentrations of the substrate should be considered. Clinical monitoring, and possibly serum digoxin levels during treatment with azithromycin and after its discontinuation is necessary.

Zidovudine
Single 1000 mg doses and multiple 1200 mg or 600 mg doses of azithromycin had little effect on the plasma pharmacokinetics or urinary excretion of zidovudine or its glucuronide metabolite. However, administration of azithromycin increased the concentrations of phosphorylated zidovudine, the clinically active metabolite, in peripheral blood mononuclear cells. The clinical significance of this finding its unclear, but it may be of benefit to patients.

Azithromycin does not interact significantly with hepatic cytochrome P450 system. It is not believed to undergo the pharmacokinetics drug interactions as seen with erythromycin and other macrolides. Hepatic cytochrome P450 induction or inactivation via cytochrome-metabolite complex does not occur with azithromycin.

Ergot
Due to the theoretical possibility of ergotism, the concurrent use of azithromycin with ergot derivatives is not recommended.

Atorvastatin
Co-administration of atorvastatin (10 mg daily) and azithromycin (500 mg daily) did not alter the plasma concentrations of atorvastatin (based on a HMG CoA-reductase inhibition assay). However, post-marketing cases of rhabdomyolysis in patients receiving azithromycin with statins have been reported.

Carbamazepine
No significant effect was observed on the plasma levels of carbamazepine or its active metabolite in patients receiving concomitant azithromycin.

Cimetidine
The effects of a single dose of cimetidine, given 2 hours before azithromycin, on the pharmacokinetics of azithromycin, no alteration of azithromycin pharmacokinetics was seen.

Coumarin-Type Oral Anticoagulants
Azithromycin did not alter the anticoagulant effect of a single 15 mg dose of warfarin administered. There have been reports received in the post-marketing period of potentiated anticoagulation subsequent to co-administration of azithromycin and coumarin-type oral anticoagulants. Although a causal relationship has not been established, consideration should be given to the frequency of monitoring prothrombin time when azithromycin is used in patients receiving coumarin-type oral anticoagulants.

Cyclosporin
Administration of a 500 mg/day oral dose of azithromycin for 3 days and then a single 10 mg/kg oral dose of cyclosporin results in cyclosporin C_{max} and AUC₀₋₅ to be significantly elevated. Consequently, caution should be exercised before considering concurrent administration of these drugs. If co-administration of these drugs is necessary, cyclosporin levels should be monitored and the dose adjusted accordingly.

Efavirenz
Co-administration of a 600 mg single dose of azithromycin and 400 mg efavirenz daily for 7 days did not result in any clinically significant pharmacokinetic interactions.

Fluconazole
Co-administration of a single dose of 1200 mg azithromycin did not alter the pharmacokinetics of a single dose of 800 mg fluconazole. Total exposure and half-life of azithromycin were unchanged by the co-administration of fluconazole, however, a clinically insignificant decrease in C_{max} (18%) of azithromycin was observed.

Indinavir
Co-administration of a single dose of 1200 mg azithromycin had no statistically significant effect on the pharmacokinetics of indinavir administered as 800 mg three times daily for 5 days.

Methylprednisolone
No significant effect on the pharmacokinetics of methylprednisolone.

Midazolam
Co-administration of azithromycin 500 mg/day for 3 days did not cause clinically significant changes in the pharmacokinetics and pharmacodynamics of a single 15 mg dose of midazolam.

Nelfinavir
Co-administration of azithromycin (1200 mg) and nelfinavir at steady state (750 mg three times daily) resulted in increased azithromycin concentrations. No clinically significant adverse effects were observed and no dose adjustment is required.

Rifabutin
Co-administration of azithromycin and rifabutin did not affect the serum concentrations of either medicinal product.

Neutropenia was observed in subjects receiving concomitant treatment of azithromycin and rifabutin. Although neutropenia has been associated with the use of rifabutin, a causal relationship to combination with azithromycin has not been established.

Sildenafil
There was no evidence of an effect of azithromycin (500 mg daily for 3 days) on the AUC and C_{max} of sildenafil or its major circulating metabolite.

Terfenadine
No evidence of an interaction between azithromycin and terfenadine. There have been rare cases reported where the possibility of such an interaction could not be entirely excluded; however, there was no specific evidence that such an interaction had occurred.

Theophylline
No evidence of interaction when azithromycin and theophylline are co-administered.

Triazolam
Co-administration of azithromycin 500 mg on Day 1 and 250 mg on Day 2 with 0.125 mg triazolam on Day 2 had no significant effect on any of the pharmacokinetic variables for triazolam compared to triazolam and placebo.

Trimethoprim/sulfamethoxazole
Co-administration of trimethoprim/sulfamethoxazole (160 mg/800 mg) for 7 days with azithromycin 1200 mg on Day 7 had no significant effect on peak concentrations, total exposure or urinary excretion of either trimethoprim or sulfamethoxazole. Azithromycin serum concentrations were similar to those seen in other studies.

PREGNANCY AND LACTATION
Pregnancy
Based on the available animal reproduction studies which performed at doses up to moderately maternally toxic dose concentrations, no evidence of harm to the fetus due to azithromycin was found. There are, however, no adequate and well-controlled studies in pregnant women. Because animal reproduction studies are not always predictive of human response, azithromycin should be used during pregnancy only if clearly needed.

Lactation
Limited information available from published literature indicates that azithromycin is present in human milk at an estimated highest median daily dose of 0.1 to 0.7 mg/kg/day. No serious adverse effects of azithromycin on the breast-fed infants were observed.

A decision must be made whether to discontinue breast-feeding or to discontinue/abstain from azithromycin therapy taking into account the benefit of breast-feeding for the child and the benefit or therapy for the woman.

Fertility
Reduced pregnancy rates were noted following administration of azithromycin. The relevance of this finding to humans is unknown.

EFFECTS ON ABILITY TO DRIVE AND USE MACHINES
There is no evidence to suggest that azithromycin may have an effect on the patient's ability to drive or operate machinery.

ADVERSE EFFECTS
Azithromycin is well tolerated with a low incidence of side effects.

Based on the available clinical trials, the following undesirable effects have been reported:

Blood and Lymphatic System Disorders
Transient episodes of mild neutropenia.

Ear and Labyrinth Disorders
Hearing impairment (including hearing loss, deafness and/or tinnitus) has been reported in some patients receiving azithromycin. Many of these have been associated with prolonged use of high doses in investigational studies. In those cases where follow-up information was available, the majority of these events were reversible.

Gastrointestinal Disorders
Nausea, vomiting, diarrhea, loose stools, abdominal discomfort (pain/cramps), and flatulence.

Hepatobiliary Disorders
Abnormal liver function.

Skin and Subcutaneous Tissue Disorders
Allergic reactions including rash and angioedema.

General Disorders and Administration Site Conditions
Local pain and inflammation at the site of infusion.

In post-marketing experience, the following additional undesirable effects have been reported:

Infections and Infestations
Moniliasis and vaginitis.

Blood and Lymphatic System Disorders
Thrombocytopenia.

Immune System Disorders
Anaphylaxis (rarely fatal).

Metabolism and Nutrition Disorders
Anorexia.

Psychiatric Disorders
Aggressive reaction, nervousness, agitation, and anxiety.

Nervous System Disorders
Dizziness, convulsions, headache, hyperactivity, hypoesthesia, paresthesia, somnolence, and syncope.

There have been rare reports of taste/smell perversion and/or loss.

Ear and Labyrinth Disorders
Deafness, tinnitus, hearing impaired, and vertigo.

Cardiac Disorders
Palpitations and arrhythmias including ventricular tachycardia have been reported. There have been rare reports of QT prolongations and torsades de pointes.

Vascular Disorders
Hypotension.
Gastrointestinal Disorders
Vomiting/diarrhea (rarely resulting in dehydration), dyspepsia, constipation, pseudomembranous colitis, pancreatitis, rare reports of tongue discoloration and infantile hypertrophic pyloric stenosis.

Hepatobiliary Disorders
Hepatitis and cholestatic jaundice have been reported, as well as rare cases of hepatic necrosis and hepatic failure, which have resulted in death.

Skin and Subcutaneous Tissue Disorders
Allergic reactions including pruritus, rash, photosensitivity, edema, urticaria, and angioedema. Frequency not known: severe cutaneous adverse reactions (SCARs) including Stevens-Johnson syndrome (SJS), toxic epidermal necrolysis (TEN), drug reaction with eosinophilia and systemic symptoms (DRESS) & acute generalised exanthematous pustulosis (AGEP).

Musculoskeletal and Connective Tissue Disorders
Arthralgia.

Renal and Urinary Disorders
Interstitial nephritis and acute renal failure.

General Disorders and Administration Site Conditions
Asthenia, fatigue, and malaise.

OVERDOSE AND TREATMENT
Adverse events experienced in higher than recommended doses were similar to those seen at normal doses. In the event of overdosage, general symptomatic and supportive measures indicated as required.

STORAGE CONDITION
Dry powder (unopen bottle): Store below 30°C.
Once reconstituted with water, store in refrigerator at 2-8°C.

INSTRUCTION FOR USE AND HANDLING
Tap the bottle to loosen the powder. Add 8 ml of water to the bottle and shake well. Shake immediately prior to use.

For children weighing less than 15 kg, the suspension should be measured as closely as possible. For children weighing 15 kg or more, the suspension should be administered using an appropriate measuring device.

SHELF LIFE
Dry powder (unopen bottle): Product should not be used beyond the expiry date imprinted on the product packaging.

Once reconstituted with water, the suspension has a shelf life of 5 days.

PACK SIZE
The powder for oral suspension is packed in HDPE bottle.
Packs of powder equivalent to 600 mg azithromycin.
Content of the bottle after reconstitution: 15 ml.

REGISTRATION NUMBER
MAL23076010AZ

CONTROLLED MEDICINE
UBAT TERKAWAL
KEEP MEDICINES OUT OF REACH OF CHILDREN
JAUIH UBAT-UBATAN DARI KANAK-KANAK

For further information, please consult your doctor or your pharmacist.

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Lt-234.00	Product Registration Holder and Manufacturer
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